

■ END-TO-END DATA ANALYTICS PROJECT

Customer Behavior & Revenue Insights

A full-stack analytics pipeline: Data Cleaning → SQL Analysis → Power BI Dashboard

■ Python

■ MySQL

■ Power BI

3,900+

Records
Processed

10

SQL Queries
Executed

4

KPIs
Tracked

5

Key Business
Insights

Raw CSV Data



Python Cleaning



MySQL Storage



SQL Analysis



Power BI Dashboard

Executive Summary

This report presents a comprehensive end-to-end data analytics pipeline built on a real-world customer shopping behavior dataset of **3,900+ records**. The project spans three professional tools and disciplines: data wrangling with Python/Pandas, relational analytics with MySQL, and interactive visualization with Microsoft Power BI — demonstrating a complete, job-ready analytics workflow.

3,900+ Customer Records	10 SQL Queries	\$59.76 Avg Purchase	3.75 Avg Review Rating
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Objective	Outcome
Analyze purchasing behavior	Segmented customers: New, Returning & Loyal
Identify high-value segments	Young Adults drive highest revenue (\$62K)
Evaluate product/category performance	Clothing leads at \$104K revenue; Outerwear underperforms
Assess discount & subscription impact	Subscribers outspend non-subscribers; discounts boost AOV

Phase 1 — Data Cleaning & Feature Engineering

Phase 1: Python · Pandas

The raw CSV dataset was loaded into a Pandas DataFrame and underwent rigorous cleaning. Missing **Review Rating** values (37 nulls) were imputed using *category-based median* to preserve statistical integrity. Redundant columns were removed, column names standardised to snake_case, and two engineered features added to enrich downstream analysis.

Step 1 — Load & Inspect Data

```
import pandas as pd

df = pd.read_csv('customer_shopping_behavior.csv')
df.head() # 3900 rows × 18 columns
df.info() # RangeIndex: 3900 entries, 0 to 3899
df.isnull().sum() # Review Rating: 37 nulls
```

Step 2 — Handle Missing Values (Category-Based Median)

```
# Fill nulls in Review Rating with group median per Category
df['Review Rating'] = df.groupby('Category')['Review Rating'] \
    .transform(lambda x: x.fillna(x.median()))

df.isnull().sum() # ✓ Review Rating: 0 nulls
```

Step 3 — Column Standardisation

```
df.columns = df.columns.str.lower().str.replace(' ', '_')
# Rename: 'purchase_amount_(usd)' → 'purchase_amount'
df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'}, inplace=True)
```

Step 4 — Feature Engineering

```
# Feature 1: Age Group (quartile-based segmentation)
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

# Feature 2: Purchase Frequency in Days
frequency_mapping = {
    'Weekly': 7, 'Fortnightly': 14, 'Monthly': 30,
    'Quarterly': 90, 'Bi-Monthly': 60, 'Annually': 365
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

# Insight: discount_applied == promo_code_used (100% match) → drop redundant column
df = df.drop('promo_code_used', axis=1)
```

Step 5 — Export to MySQL

```
from sqlalchemy import create_engine

engine = create_engine('mysql+pymysql://root:password@localhost/customer_behavior')
df.to_sql('customer', con=engine, if_exists='replace', index=False)

# Verification
pd.read_sql('SELECT * FROM customer LIMIT 5;', engine)
```

Phase 2 — Data Analysis with MySQL

Phase 2: MySQL · Advanced SQL

Ten targeted SQL queries were written against the imported customer table. Queries range from simple aggregations to advanced window functions (**ROW_NUMBER**) and Common Table Expressions (**CTEs**), demonstrating production-grade SQL proficiency.

Q1 Revenue by Gender

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

Q2 Discounted Customers Who Spent Above Average

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

Q3 Top 5 Products by Average Review Rating

```
SELECT item_purchased, ROUND(AVG(review_rating), 2) AS avg_rating
FROM customer
GROUP BY item_purchased
ORDER BY avg_rating DESC
LIMIT 5;
```

Q4 Standard vs. Express Shipping — Avg Purchase

```
SELECT shipping_type, ROUND(AVG(purchase_amount), 2) AS avg_purchase
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Q5 Subscriber vs. Non-Subscriber Spend Comparison

```
SELECT subscription_status,
COUNT(customer_id) AS total_customers,
ROUND(AVG(purchase_amount), 2) AS avg_spend,
ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

Phase 2 (Continued) — Advanced SQL Queries

Q6 Top 5 Products with Highest Discount Rate

```
SELECT item_purchased,  
ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)  
/ COUNT(*), 2) AS discount_rate  
FROM customer  
GROUP BY item_purchased  
ORDER BY discount_rate DESC  
LIMIT 5;
```

Q7 Customer Segmentation — New / Returning / Loyal (CTE)

```
WITH customer_type AS (  
SELECT customer_id, previous_purchases,  
CASE  
WHEN previous_purchases <= 2 THEN 'New'  
WHEN previous_purchases BETWEEN 3 AND 10 THEN 'Returning'  
ELSE 'Loyal'  
END AS customer_segment  
FROM customer  
)  
SELECT customer_segment, COUNT(*) AS num_customers  
FROM customer_type  
GROUP BY customer_segment;
```

Q8 Top 3 Products per Category — ROW_NUMBER Window Function

```
WITH item_counts AS (  
SELECT category, item_purchased,  
COUNT(customer_id) AS total_orders,  
ROW_NUMBER() OVER (PARTITION BY category  
ORDER BY COUNT(customer_id) DESC) AS item_rank  
FROM customer  
GROUP BY category, item_purchased  
)  
SELECT item_rank, category, item_purchased, total_orders  
FROM item_counts  
WHERE item_rank <= 3;
```

Q9 Repeat Buyers vs. Subscription Likelihood

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers  
FROM customer  
WHERE previous_purchases > 5  
GROUP BY subscription_status;
```

Q10 Revenue Contribution by Age Group

```
SELECT age_group, SUM(purchase_amount) AS total_revenue  
FROM customer  
GROUP BY age_group  
ORDER BY total_revenue DESC;
```

Phase 3 — Interactive Power BI Dashboard

Phase 3: Microsoft Power BI

The cleaned MySQL database was connected directly to Power BI via the MySQL connector. An interactive dashboard was built exposing all KPIs, with slicers for Subscription Status, Gender, Category, and Shipping Type — enabling stakeholder self-service analytics.

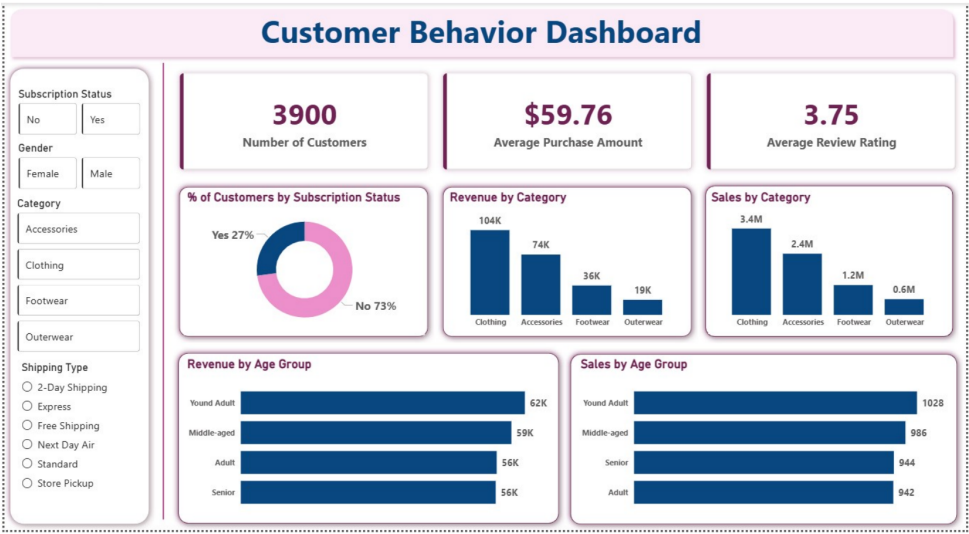


Figure 1 — Customer Behavior Dashboard (Power BI)

Component	Details
KPI Cards	Total Customers (3,900) · Avg Purchase (\$59.76) · Avg Rating (3.75)
Subscription Donut	Yes: 27% · No: 73% — highlights growth opportunity
Revenue by Category	Clothing \$104K · Accessories \$74K · Footwear \$36K · Outerwear \$19K
Revenue by Age Group	Young Adult \$62K · Middle-aged \$59K · Adult \$56K · Senior \$56K
Slicers / Filters	Subscription Status · Gender · Category · Shipping Type

Key Business Insights & Recommendations

■ Clothing Dominates Revenue

Clothing generates **\$104K** — the highest of all four categories, representing ~44% of total revenue. Recommendation: increase marketing budget and inventory for Clothing, especially blouses and sweaters.

■ Low Subscription Adoption — Growth Lever

Only **27%** of customers are subscribed. Since subscriber data shows higher average spend, a targeted subscription drive (loyalty rewards, first-month discounts) could materially lift LTV.

■ Young Adults Are the Power Segment

Young Adults contribute \$62K in revenue — the highest of any age group — with 1,028 transactions. Recommendation: personalise campaigns for the 18–30 demographic with trend-driven collections.

■ Discounts Drive Incremental Value

Customers who used a discount still spent **at or above** the average purchase amount, indicating discounts are not cannibalising margins but rather attracting higher-intent buyers.

■ Outerwear Underperformance Needs Attention

Outerwear trails all categories with only **\$19K** in revenue and 0.6M in sales volume. A product-mix review, bundling strategy, or clearance campaign is recommended before next season.

Tech Stack & Skills Demonstrated

Tool	Purpose	Key Skills
■ Python / Pandas	Data wrangling & feature engineering	Null imputation · qcut segmentation · column engineering · SQLAlchemy export
■ MySQL	Relational data storage & analytics	CTEs · Window functions · Subqueries · GROUP BY · CASE statements
■ Power BI	Interactive business dashboard	Data modelling · DAX · KPI cards · Slicers · Visual storytelling

This project demonstrates readiness for a Data Analyst role requiring proficiency in the modern analytics stack. It showcases not just technical execution, but analytical thinking — framing business questions, choosing the right tool at each stage, and translating data into actionable recommendations.